ARCHITECTURE OVERVIEW

The Inaya Ecosystem, End to End

How the storage protocol, the applications built on it, and the AI layer all fit together — the shape of the system, without the code.

Document INAYA-OV-2026-V1 · Classification Internal · October 2026

SECTION 01

The Shape Of The System

Inaya is one ecosystem built on two connected layers. The bottom layer is a real DePIN protocol — people encrypt their own files, independent operators store and verify them, and a blockchain anchors who owns what. The top layer is a set of applications — web, mobile, two desktop apps, and a business SaaS product — that most users interact with without ever knowing a blockchain is involved.

THE PROTOCOL LAYER

Smart contracts that handle ownership, staking, node payouts, and threat verification. This is where the token economics live, and where independent node operators actually earn.

THE APPLICATION LAYER

Everything a person actually opens: the storage dApp, the Business Workspace for companies (including its own operations, finance, and HR tooling), the mobile app, two desktop apps, a public security page, an investor data room, and four AI assistants — all talking to the same backend.

SECTION 02

Storage — How A File Actually Gets Protected

This is the core promise: nobody but the file's owner can ever read it — not Inaya, not the node storing it, nobody.

1. **Encrypt on your device.** Before a file ever leaves the user's browser or phone, it's encrypted using a key derived from their own passkey. That passkey is never sent anywhere.
2. **Split it in two.** The encrypted file is split into two halves. Neither half means anything on its own — you need both to reconstruct anything.
3. **Store both halves.** The two halves get pinned to decentralized storage (IPFS/Pinata), independent of each other.
4. **Anchor ownership on-chain.** A record — who owns this file, where its two halves live, a fingerprint proving it hasn't been tampered with — gets written to the blockchain. This is the permanent, tamper-evident receipt.
5. **Reverse it to read.** Downloading does the same thing backwards: look up the on-chain record, fetch both halves, put them back together, decrypt with the owner's passkey. Nothing about this process gives Inaya, a node operator, or anyone else a way to see the plaintext.

> **You don't have to take Inaya's word for any of this.** The encryption code above is published as open source (@inaya-network/custody-sdk on npm) and used identically by the web app and the mobile app — not a separate, unverifiable implementation behind the scenes. Every release publishes a cryptographic hash anyone can independently reproduce and check, plus a second copy pinned to IPFS so the code itself, not just Inaya's word, identifies what you're running. Full verification steps are public.

> **The two halves don't depend on one company staying up.** Each file's two encrypted shards are replicated across two independent pinning providers on genuinely different infrastructure (Pinata and Filebase), not just one. The system checks pin health continuously and, if a replica goes missing or its content stops matching what was originally captured, automatically fetches from the surviving healthy copy, re-verifies it, and re-pins — self-healing back to fully protected without anyone having to notice or act.

SECTION 03

Node Operators — Who Actually Stores The Data

Independent people and organizations run node software that registers them on-chain and reports their capacity. In exchange, they earn USDT and $INAYA — for storage service, and separately, for contributing to a decentralized threat-intelligence network (see Section 05).

- Registration and identity live on-chain — a node's stake, tier, and track record are public and verifiable.
- Payouts are deliberately delayed and verifier-attested rather than instant and self-reported — this protects the system from a single bad actor (or a single compromised key) draining funds in one move.
- Today, the software an operator runs handles registration and reporting — it doesn't yet handle the actual physical hosting of file shards. That's a known, stated gap in the current build, not a hidden one.

SECTION 04

The Web dApp — Where Consumers And The Protocol Meet

The main website is where an individual user does everything token-and-storage related: gets testnet tokens from a faucet, uploads/downloads files in the Sovereign Vault, stakes $INAYA for yield, tracks their activity on a dashboard, refers friends (email + identity verification, no wallet needed for that part), and — for larger customers — buys an annual Corporate Reserve storage plan, paid in USDT and settled on-chain.

> Staking has real lock-period tiers (flexible, 30-day, 90-day) with better rewards for longer commitments — this is enforced on-chain, not just a UI label.

SECTION 05

Business Workspace — A Second Product On The Same Foundation

Not every customer wants a blockchain product. Business Workspace is a genuine B2B SaaS tool — organizations, departments, projects, documents — with real approval workflows, granular permissions, secure sharing links, and a permission-aware AI assistant. Sign-in is just an email or a Google account. No wallet, no tokens, no blockchain vocabulary anywhere in the experience.

It has grown well beyond documents: real business operations (Tasks, CRM, Procurement, Inventory) and a Finance & HR layer (invoicing, expense approval, employee records, leave management) now run on the exact same organization/department/permission foundation described above — every new module additive, nothing about the base system had to be restructured to add them. Finance & HR is explicitly a testnet demonstration layer today, not regulated banking or payroll.

> **Why this matters strategically.** It's a second, independent revenue engine — priced and sold like any other business software, evaluated by a buyer who will never ask what chain it runs on — while still running on top of the same infrastructure and the same encryption guarantees as the consumer storage product.

SECTION 06

Security Layer — A Decentralized Threat Network

Marketed publicly as "Inaya Firewall." The same trust model as the storage layer, applied to threat intelligence instead of files: independent nodes report suspicious websites, the network only trusts a threat once enough independent, reputable reporters agree, and confirmed verdicts get permanently recorded on-chain so they can't quietly be changed later.

- A public website page lets anyone — no account needed — check a domain, see network stats, and ask an AI assistant about how it all works.
- The mobile app can check destinations before you trust them and keep your own personal block/allow list.
- The desktop apps can, on Windows, actually block a confirmed-malicious address at the operating-system firewall level, not just inside the app window.

SECTION 07

The AI Layer — Four Assistants, One Pattern, Different Rules

All four assistants (Docs, Business, Security, Learn Tutor) are built the same technical way — they can call small, purpose-built tools to look up real data before answering. What's different is the philosophy each one follows.

ASSISTANT	ITS ONE RULE
Docs Assistant	Answer grounded in the project's own real documentation — retrieves the relevant passage first, then answers from it, and cites where the answer came from.
Business Assistant	Never show a user data they aren't actually permitted to see — every answer respects real document/project/finance/HR permissions.
Security Assistant	Never invent a threat verdict — every specific claim must be backed by real, verified network data.
Learn AI Tutor	The opposite of the other two — teach freely using its own knowledge, like a real tutor would; only use tools to check the user's own saved videos and progress.

> **How the Docs, Security, and Learn assistants find the right passage.** A shared retrieval layer (RAG) indexes the project's real documentation and FAQs, embeds it for semantic search, and combines that with plain keyword search — so "how does the pricing work" and "pricing structure" both find the same real answer, not a static block of hardcoded text that goes stale the moment the product changes.

AI-Powered Business Operations — the AI can propose, it can never execute.

As of September 2026, the Business Assistant can go further than answering questions: it can propose real changes across 9 business domains (Tasks, Expenses, Documents, Employees, Invoices, Leave Requests, Purchase Orders, Purchase Requests, Deals). It never performs any of them itself.

1. **The AI proposes, not executes.** Every proposal is checked against the exact same permission the real action would require before it's even created — the AI can't propose something the requesting user isn't already allowed to do.
2. **A human approves with real authority.** A person with the exact same real authority the underlying action requires — re-checked independently by the server, not trusted from the request — must approve it. Higher-risk actions (terminating an employee, approving a large purchase) are automatically classified as higher risk than routine ones (advancing a task).
3. **A mandatory 36-hour delay passes.** Even after approval, nothing executes immediately — mirroring the same timelocked-settlement pattern the storage protocol itself uses for node-operator payouts, reused here for a completely different kind of action.

4. **Only then does it execute — and everything is recorded.** A separate process executes the change only once the delay has genuinely passed, and only if the real action still makes sense. Every step — proposed, approved, delayed, executed, rejected, or expired — is written into a cryptographically verifiable audit trail.

> **The audit trail isn't just a log — it's tamper-evident by construction.** Every entry cryptographically commits to the one before it, so altering or deleting any past entry breaks every entry after it — a direct database edit is mathematically detectable, not just against policy. Any business customer can independently walk their own organization's chain and recompute it themselves via a self-service export; they don't have to trust a "Verified" badge Inaya shows them. Security-tested with 19 automated tests, 11 of them adversarial scenarios specifically trying to defeat the guardrails.

SECTION 08

Mobile & Desktop — Same Backend, Different Doors

None of the client apps are separate products with their own logic — they're different doors into the exact same backend and the exact same data.

- The mobile app is a full superset — everything the web dApp does, plus the Business Workspace, Learn, and Security, in one app.
- The two desktop apps are deliberately thin — each one is just the real website, running inside a lightweight native window with a system tray icon, native notifications, and auto-updates layered on top. There is no separate desktop-only feature set to keep in sync.

SECTION 09

App Store, NFT Vault & Account Security (September 2026)

Three smaller but real additions rounding out the consumer side: a place to discover Web3 apps, a place to back up NFTs the same way files are backed up, and a stronger login.

- Web3 App Store (/apps) — a curated set of Inaya's own apps, plus community-submitted ones. A submission needs a wallet signature and is checked against the live threat network before an admin ever sees it, and nothing goes live until approved.
- NFT Vault (/nfts) — discovers NFTs a wallet owns and backs up their metadata and image using the exact same encrypt-and-shard pipeline that protects every other file on Inaya.
- Optional two-factor login for Business Workspace — an authenticator app or a text message code, on top of the existing passwordless sign-in.

SECTION 10

Multi-Chain — Moving Value Beyond BNB Chain

Inaya started on BNB Chain Testnet, but $INAYA doesn't have to stay there. Two independent systems, running in parallel, let value move to other chains — deliberately kept separate, since a problem in one says nothing about the other.

INAYA'S OWN NATIVE BRIDGE

Full control, no third-party dependency. A real, proven lock-and-mint cycle now works to Ethereum, Avalanche, Arbitrum, Solana, Hedera, Aptos, and Sui — each one verified with a real transaction that moved real (testnet) value and confirmed it arrived on the other side, not just deployed-and-assumed-working.

A SECOND ROUTE VIA WORMHOLE

An independent, third-party interoperability network, integrated as an alternative path that doesn't replace the native bridge. Proven working end-to-end to Ethereum, Arbitrum, and Avalanche; a few other chains are currently blocked by issues in Wormhole's own tooling, not anything on Inaya's side, and are shown as reference-only until that's resolved upstream.

Nothing here is claimed beyond what's actually been proven on-chain. A chain only ever gets credit for the capability level it's genuinely demonstrated — discovered, wallet-connected, messaging-only, or a real proven transfer — never assumed from "it's deployed" alone. Several real bugs were found and fixed getting the newer chains (Solana, Sui) working; each is documented rather than smoothed over, the same honesty convention this whole ecosystem holds itself to.

SECTION 11

Oracle & Automation — Infrastructure That Runs Itself

A third independent subsystem: an on-chain registry of approved data sources, and a keeper that executes pre-approved contract actions automatically — under smart-contract rules, never arbitrary admin commands. Live on BSC Testnet today, publicly verifiable at inayanetwork.com/automation.

- The live demo isn't simulated: the price feed reads the real PancakeSwap testnet pool for INAYA/USDT, and the automated action is a real, already-existing node-settlement release function on InayaNodeRegistry — the first run found and released an actual previously-unclaimed payout.
- If the data feeding an automated action goes stale, the system skips that action rather than acting on unverified information — proven live, not just claimed.
- The keeper never holds broad authority — it can only submit data it's explicitly registered for, and only call functions that are already safe for anyone to call.

SECTION 12

Sovereign Enterprise OS — One Operating Environment, Not a Pile of Features

Almost everything described in this document already existed as a separate feature. What was missing was the connecting layer — one place that shows you what needs your attention, lets you search everything at once, asks one assistant instead of four, and tells you plainly whether your data and your organization's activity are in a trustworthy state right now.

- One OS Home screen — trust & health status, recent notifications, what changed this week, and an assistant that can answer both business and security questions — on both the web app and Business Workspace, each tailored to how you actually use it.
- Real notifications with a memory — mark something read once, and it stays read, instead of the app forgetting the moment you refresh.
- One search bar (Cmd/Ctrl+K) instead of hunting through a dozen separate screens for a task, a deal, a document, or a file.
- A trust & health signal that's actually true, not decorative — built from the same real audit trail, backup health, and security data described earlier in this document, never invented for the sake of a clean-looking dashboard.
- On desktop, a module you're watching closely (like the audit trail) can now pop out into its own window and stay visible alongside everything else, instead of switching away from what you were doing.

> **Two operating environments, not one blended one.** Your business identity (email, organization membership) and your wallet identity are kept genuinely separate — there was no clean way to merge them without inventing a new identity system neither product actually needed, so the OS shell was built twice from the same underlying pieces instead: one for Business Workspace, one for the wallet-connected app. This mirrors how the two desktop apps already work today.

SECTION 13

Healthcare & Legal OS — Two Industries, Same Secure Foundation

Business Workspace now has two specialized modes for organizations with genuinely different needs: Health OS for hospitals and clinics, Legal OS for law firms and legal departments. Both are built on the exact same encryption, audit trail, and permission model as the rest of Business Workspace — just adapted to the records those two industries actually work with.

- Need-to-know access, not just department-wide — only staff actually assigned to a patient's care team, or a lawyer actually on a matter's team, can see that record. Being in the right department isn't enough on its own, which is stricter than how the rest of Business Workspace works, on purpose.
- Emergency access that's actually safe — a clinician can get emergency access to a patient outside their normal assignment, but it's logged the instant it happens, expires automatically after a few hours, and a manager is notified to review it. We caught and fixed a real bug during testing where an early version didn't expire on time.
- A legal hold that actually holds — placing a hold on a matter doesn't just show a warning, it genuinely blocks anyone from deleting the related records until the hold is released.
- Trust accounting with one hard rule — the system will never let anyone withdraw more from a client's trust fund than what's actually there, no matter who's asking.
- An AI assistant that knows its limits — it can summarize, search, and draft, but it's built to flatly refuse anything that crosses a real line: diagnosing a patient, giving final legal advice, or agreeing to release a hold or delete evidence.
- The right modules for the right business — a general company never sees Health OS or Legal OS at all; a hospital or law firm picks its type at signup (or changes it later in Settings), and that choice is enforced for real, not just hidden behind a menu a determined user could still reach.

> **What's live today vs. what's next.** Patient records and legal matters both have a real, working screen in Business Workspace today. Most of the other modules described above — consent, billing, scheduling, evidence, legal holds, contracts, trust accounting — work correctly behind the scenes and are already tested, but don't have their own screen in the app yet; that's the next phase of work, building on logic that's already written rather than starting over. Nothing here is HIPAA-certified, ABA-certified, or legally compliant with any specific jurisdiction — that would require a real legal/compliance review this build doesn't include, and we've been careful not to claim otherwise.

SECTION 14

Financial Services, Regulated Enterprise & Government — Three More Industries, Same Foundation

Three more specialized modes now sit alongside Health OS and Legal OS, all sharing the exact same encryption, audit trail, and permission model: Financial Services OS for hedge funds and asset managers, Private Capital OS for private equity and venture capital, Regulated Enterprise OS for banks, insurers, and other heavily-regulated companies, and Government OS for public-sector departments and agencies.

- A compliance foundation that never overstates itself — controls can be mapped to real industry frameworks (NIST, ISO 27001, SOC 2, DORA, GDPR), but every mapping is clearly labeled a reference, never a certification. A control nobody has tested yet shows up as 'unknown,' never a falsely reassuring green checkmark.
- A published policy can't be quietly rewritten — changing it always creates a brand-new version, so there's always a clear record of what a rule said on any given day. The Government OS Policy Knowledge Base reuses this exact same rule.
- Real fund, deal, and investor tracking for the finance side — research, investment committee decisions, portfolios, and for private equity/VC: deal pipelines, due diligence, cap tables, and portfolio-company boards.
- Government's need-to-know rule, borrowed from Health OS — a caseworker with general staff access still can't open a specific citizen's record unless they're actually assigned to it. Every citizen-record read is logged, not just every change — a stricter bar than the rest of Business Workspace.
- An outside examiner or auditor gets their own scoped, one-time-use link — never a real login to the company's system — for both regulatory examinations and Government OS's own case reviews.
- AI assistants that know their limits here too — they can search, summarize, and draft, but are built to flatly refuse a request to certify compliance, approve something final, or override a permission, on any of these four verticals.

> **What's live today vs. what's next.** Financial Services OS, Private Capital OS, and Regulated Enterprise OS are fully built on the web app, verified end-to-end. Government OS's core (citizen records, case management, the policy knowledge base, the operations/security dashboard, and its own AI assistant) is also built and tested on web. None of the four have a mobile screen yet — that's next, building on logic that's already written. No certification, accreditation, or jurisdiction-specific regulatory approval of any kind is claimed for any of these — that requires independent authorities this build doesn't include, and we've been careful not to claim otherwise.

SECTION 15

Trust You Can Check Yourself, and Recovery That's Actually Re-Tested

Two additions that both answer the same question a different way: not 'trust us,' but 'here's how you check.'

- A public Trust Center (inayanetwork.com/trust) that anyone can visit without logging in — and its centerpiece isn't a badge that just says 'Verified.' It recomputes the cryptographic proof itself, in your own browser, and tells you exactly where a record breaks if it ever does.
- We found a real bug in that very verifier while testing it against a real record: it originally assumed every proof started from the very beginning of the chain, so a smaller, single-record proof was wrongly flagged as broken. Fixed, and now the page is honest about which of the two proof strengths you're actually looking at.
- A real way for two organizations to formally trust each other — propose, accept, revoke — though today it's the foundation for that, not yet a feature anything else in the app uses to actually share data.
- Developer API keys that always act as the organization that created them, never anyone else's — down to the design, not just the policy.
- Recovery testing that actually runs, automatically, against real systems — using safe stand-in files Inaya owns itself, since testing with a real customer's files would require a password Inaya was never given in the first place. Every test's real pass/fail result becomes part of the same tamper-evident record everything else in this document already uses.

> **What this doesn't claim.** No SOC 2, HIPAA, ABA, or FedRAMP certification is claimed on the Trust Center — it says so directly. And a specific customer's own resilience-test results stay private to that customer, never published on the public page.

SECTION 16

A Real Home for Node Operators

The people who actually run Inaya's storage nodes now have their own dashboard — separate login, separate from the wallet app, separate from Business Workspace — showing uptime, rewards, and network standing.

- Real uptime history, not guessed — and if the dashboard doesn't yet have enough history to answer a question honestly (say, 90 days of data on day one), it says so instead of estimating.
- Real numbers read directly from the blockchain for the things that matter most — an operator's tier and earnings — not a copy that could go stale.
- There's deliberately no 'release my payment' button — payouts are already fully automatic on a timer nothing in this dashboard can shortcut; it can only show you where things stand.

SECTION 17

Four New Ways to Prove and Protect Business Value

Four new tools in Business Workspace, each doing something genuinely useful — described precisely, so nobody mistakes what's cryptographically guaranteed for what's a well-built internal workflow.

- Inaya Sign — document signing with a real digital signature bound to the exact document and moment it was signed; if the document changes afterward, that signature stops being valid for it. A completed signing can optionally be stamped onto the blockchain as extra proof, on top of the signature itself.
- DePIN Storage Manager — lets a business register and monitor its own storage nodes today; it doesn't yet route actual files through them the way the public node network does, and we say so plainly in the product rather than implying it already does.
- Milestone Escrow — a real, enforced 'don't release payment until a milestone is confirmed' workflow, with mandatory human approval and a built-in waiting period before money actually moves. It's an internal safeguard, not a blockchain escrow contract holding funds — we describe it that way on purpose.
- Financial Attestation — a business can cryptographically prove a claim about its own finances (like 'revenue exceeded $X this quarter') without a third party needing to see every underlying transaction, using a real cryptographic fingerprint that changes if a single number is later altered. It's a strong, real guarantee — just not the specific 'zero-knowledge proof' technique, which this doesn't use.

SECTION 18

Smarter Day-to-Day Operations

Real improvements to the daily business tools, plus an AI assistant that guides rather than acts.

- Invoices and purchase orders now support five currencies, always showing the original amount and a converted figure side by side, using a clearly dated reference rate rather than a live market feed.
- One-click professional invoices, moving stock between warehouses with a full paper trail, editable purchase orders before they ship, and new downloadable reports.
- An AI assistant that can now walk someone through a multi-step task, one real step at a time — but it never performs the step for you. It only advances once it sees real proof you actually did it, not just because you told it you did.

SECTION 19

Real Connections to the Tools You Already Use — and a Voice to Talk To

Two upgrades to how people actually interact with the platform day to day.

- Nine business integrations (accounting, communication, storage, and productivity tools) now connect for real — a genuine sign-in with each provider, not a placeholder 'Connect' button. Disconnecting one now genuinely revokes access, instead of just hiding a status.
- The Business AI Assistant can now be spoken to and talks back — the same assistant, the same limits, the same inability to take a real action without a real person's approval. Voice is a new way to reach it, not a new set of permissions.

SECTION 20

A Workspace That's Easier to Actually Use

A real usability pass across Business Workspace — not a visual redesign, since the look was already consistent. The actual problem: the same interface pieces had been hand-copied dozens of times instead of built once and shared, and a few navigation choices genuinely confused people.

- Two competing 'home' screens became one. Clicking Departments, Projects, or Documents used to all land on a screen generically titled 'Company Records' no matter which you clicked — now it correctly shows which one you're in.
- Every form field got a real, readable label for the first time — previously, faint placeholder text was the only clue what a field was for.
- None of this touched how permissions, security, or any backend logic works — it's exclusively about making the same real functionality easier to find and use.

SECTION 21

Multi-Cloud Storage Compatibility & Enterprise Storage Governance (September 2026)

The storage layer now speaks the two protocols enterprise IT already standardizes on — a real AWS S3-compatible API and a real Azure Blob Storage-compatible API, each verified against the actual AWS CLI and the official Azure SDK, not a simplified approximation. An organization's existing scripts, backup tools, and SDKs connect directly, with no migration and no new tooling to install.

- Scoped access credentials — a credential can be restricted to one bucket, one folder path, a specific set of operations (read/write/delete/list), and an expiry date, enforced server-side on every request.
- Object versioning — overwriting a file keeps prior versions retrievable and restorable, opt-in per bucket.
- Object Lock — immutable, time-locked retention for compliance, enforced identically whether the request comes through the S3 API, the Azure API, or Business Workspace.
- Legal hold — an indefinite deletion block for litigation/investigation holds, independent of any retention timer.
- Lifecycle policies — automated expiration by age, with locked or legally-held objects always exempt.
- Inaya Drive — a real Windows drive letter backed by the same storage, live-tested end to end: mount, browse, open, save, and delete files the same way you would on any local or network drive.

How this fits the zero-knowledge model described in Section 02. Standard S3/Azure client tools were never built to encrypt a file client-side before sending it — that step doesn't exist in the AWS CLI or the Azure SDK. So objects written through this compatibility layer specifically use a server-managed encryption key, scoped per organization, itself encrypted at rest, with every use logged to that organization's own audit trail — the same trust model any real S3-compatible gateway product operates under. This is a disclosed, structurally necessary exception for this one access path; every other Inaya document — uploaded through the web app, mobile app, or Business Workspace — keeps the client-side-only model Section 02 describes, unchanged.

Inaya Drive's WinFSP-backed filesystem code runs as its own separate process (inaya-drive-helper), not linked into the desktop app's own binary — a deliberate licensing boundary, since the real Rust WinFSP binding is GPL-3.0 and this keeps that license off Inaya's proprietary desktop app entirely. Windows-only at the time this section was written — Section 23 below covers Linux support added since.

SECTION 22

How It All Actually Connects

One backend serves every surface — the website, the mobile app, and both desktop apps all call the exact same API. There's no duplicated logic to keep in sync between platforms; a fix or feature on the backend is instantly live everywhere.

The protocol (contracts, node operators, encryption) and the applications (dApp, Business Workspace, mobile, desktop, AI) are cleanly separated — most of what a user touches day-to-day never has to think about the blockchain underneath it, even though it's doing real work a layer down.

BNB Chain Testnet is still home base — the core protocol, staking, and node settlement all live there — but it's no longer the only chain involved: $INAYA now moves to several other testnets through the bridge described in Section 10. Nothing described in this document is live on mainnet yet, on any chain — that's the next major milestone, not a past one.

SECTION 23

Removing the Last Reasons Not to Switch (September 2026)

The gap between "Inaya is technically compatible with S3/Azure/GCS" and "a real enterprise can actually move over without friction" closed further this month: existing data can now be migrated in directly, popular backup and infrastructure tools were tested against Inaya for real, and three genuine gaps enterprise buyers would have hit immediately — empty folders, Google account sign-in, and shareable download links — are now closed.

- A local Data Migration Agent moves existing AWS S3, Azure Blob, or Google Cloud Storage data directly into Inaya — every credential stays on the customer's own machine, never touching Inaya's servers or browser. Interrupted runs resume safely; nothing is ever silently duplicated.
- Real backup/infrastructure tool compatibility, proven rather than assumed: rclone and Terraform both verified fully working end to end against the live endpoint, including Terraform's own full provision-and-tear-down lifecycle.
- Inaya Drive now mounts on Linux as well as Windows, with real empty-folder creation on both — the same New Folder action any normal drive supports, proven to survive killing and restarting the mount process entirely.
- Sign in to the storage endpoint directly with a Google account, mapped to existing organization permissions — no separate Google-only access tier.
- Temporary, time-limited download links for sharing a single file without handing out a password — with a real, measured expiration, not a promised one.
- A one-click Compliance Evidence Exporter — a downloadable, cryptographically-hashed evidence package built entirely from records Inaya already keeps, for a customer's own audit or legal process.

> **An honest correction, not just a new feature.** An earlier finding that Google's own "gcloud storage" tool couldn't connect to a third-party endpoint like Inaya's turned out to be incomplete — it can, through an official (if Google-labeled "unstable") configuration option, and real uploads and downloads were proven working. The older "gsutil" tool still can't, due to a genuine bug in Google's own software that a workaround narrowed down further but didn't fully resolve — reported plainly rather than claimed fixed.

> macOS support for Inaya Drive is written but not yet tested on real Mac hardware, and is not claimed as supported until it is. Full writeup: docs/inaya-drive-empty-folder-creation-report.md, docs/enterprise-adoption-and-market-reach-report.md, and docs/gcs-compatibility-extension-report.md.

SECTION 24

From Storage to Trust: Evidence, Simulation, and a Rounder Storage Feature Set (September 2026)

Two things happened this month that push Inaya further past "technically compatible" and toward "a platform a company can actually build trust workflows on": storage picked up the remaining features an IT team expects (labels, an inventory report, bulk actions, a security checkup), and a new layer connects invoices, purchase orders, and AI decisions into one provable story — plus the ability to ask "what would happen if..." about the business itself, safely, before committing to anything real.

- Storage objects can now be labeled (tagged), given a real independently-verifiable fingerprint, exported as a one-click inventory report, and updated in bulk across up to 1,000 files at once — all while Object Lock and legal holds remain fully enforced, never bypassed by a bulk action.
- A new Business Event links an invoice, purchase order, or AI-proposed action to everything Inaya checked about it — with a one-click "Why?" explanation and a portable, cryptographically-provable passport an auditor or partner can independently verify.
- A "What If?" simulator answers questions like "what happens if I approve this purchase?" or "what happens if a supplier becomes unavailable?" using the company's own real, connected data — and is proven, by automated test, to never actually change anything just by asking.
- The same simulation extends into a lightweight Digital Twin: supplier unavailable, employee loses project access, project delayed, or warehouse unavailable — each answer is honest about what's genuinely unknown rather than guessing.

> **An honest correction, not just a new feature.** A genuine, previously-unnoticed misconfiguration was found and fixed during this work: the intended backup storage provider (Pinata) had a broken credential and had never actually been receiving backup data — the platform was silently relying on its secondary provider as if it were primary. Now fixed and verified with a real, live round trip.

> Added in October 2026: storage event notifications, which send a signed message to a customer's own endpoint when files are uploaded, deleted or expire. Deliberately not built yet: connecting two different companies' Digital Twins to each other privately (would require heavyweight cryptography not yet justified by a real customer need — researched and documented, ready for when it is). Full writeup: docs/aws-s3-feature-expansion-report.md, docs/evidence-graph-business-event-layer-report.md, and docs/digital-twin-simulation-layer-report.md.

SECTION

Modular Enterprise Adoption Layer — DirectSync, Data Rooms, Cloud Backup & What-If Studio (September 2026)

Four self-contained features, built after a mandatory audit of what already existed, so nothing here is a second version of a system Inaya already had: automatic local-folder backup, ready-made secure data-room templates, a recurring cloud-to-Inaya backup scheduler, and a visual front-end over the existing Digital Twin simulator.

- DirectSync — point it at a folder on your computer and it automatically, quietly backs it up into Inaya in the background. Duplicate-safe (it never re-uploads an unchanged file), resumable if the connection drops, and a local delete never deletes the backup — that's a deliberate safety choice, not an oversight.
- Data Room Templates — four ready-made secure sharing setups (Fundraising, M&A, Legal Review, Web3 Due-Diligence) so a confidential document room with the right structure, NDA gate, and access rules can be created in seconds instead of built by hand each time.
- Cloud Backup Scheduler — set it once, and a company's existing AWS, Azure, or Google Cloud storage stays automatically mirrored into Inaya on a recurring schedule, only re-copying what actually changed, with a simple Healthy/Warning/Degraded/Failed status at a glance.
- What-If Scenario Studio — a proper interface over the existing "what would happen if..." simulator, with a history of past scenarios and a clear before-and-after comparison, so a manager doesn't need to know an API exists to use it.

> **Tested for real, not assumed to work.** DirectSync's core promises — no duplicate uploads, a rename doesn't create a second copy, a local delete never touches the backup — were each proven by running a real end-to-end test against a real running server and real files on a real Windows computer. That same test caught a genuine bug (a rename wasn't actually relocating the backed-up file) before this was called finished.

> Honestly scoped: DirectSync currently runs on Windows, tested for real; the Linux version uses the same underlying pieces already proven to work on Linux elsewhere in Inaya, but hasn't itself been run on a real Linux machine yet, so it's marked unverified rather than claimed. Full writeup: docs/directsync-report.md, docs/cloud-backup-scheduler-report.md, and docs/MODULAR_ADOPTION_CAPABILITY_AUDIT.md.

SECTION

A Storage Control Plane and a Real Terraform Provider (September 2026)

Inspired by IBM Cloud's own VPC storage product line: a proper control panel for a business's own Inaya storage — volumes, file shares, point-in-time snapshots, automated backup schedules — plus, once real developer tooling was set up, a genuine Terraform provider so an IT team can manage all of it as code the same way they already manage their other cloud infrastructure.

- Storage volumes and file shares — a single, organized place to create, tag, resize, and track a company's storage, each one backed by a real Inaya storage bucket underneath.
- Real point-in-time snapshots — genuinely incremental (a snapshot only stores what changed, not a full copy every time), restorable, and shareable with another organization on a revocable, expiring basis.
- Automated backup policies — pick which storage to protect by label, set a daily/weekly/monthly/long-term schedule and how many backups to keep, and Inaya handles the rest automatically, including cleaning up old backups.
- A real Terraform plugin — the same kind of automation tool IT teams already use for AWS or Azure now works for Inaya's storage too, letting infrastructure be defined and version-controlled as code instead of clicked through a website.

> **The one honest limit, stated plainly.** IBM's "attachable hard drive" feature assumes a running virtual computer to plug it into — Inaya doesn't run virtual computers for customers, so that specific piece is a real, useful control-panel substitute (reservations and organized tracking) rather than a literal physical disk. Every storage resource states this plainly rather than implying more than it delivers.

> Tested for real: 35 automated tests against the real database and real storage system, plus the new Terraform plugin's full create-change-delete cycle run against a real running copy of Inaya, not a dry run — a real bug (blank status information right after creating something) was caught and fixed along the way. The Terraform plugin isn't published to Terraform's public directory yet; that's a small follow-up step, not more engineering. Full writeup: docs/ibm-vpc-storage-expansion-report.md and terraform-provider-inaya/README.md.

SECTION

Official Documentation Platform — inayanetwork.com/docs (September 2026)

A real, official documentation site, in the spirit of IBM Cloud's own docs portal: product guides, a full API/SDK/CLI reference, search, release notes, and a downloadable machine-readable spec — every claim on it verified against the actual product, not aspirational copy.

- Product guides, written in plain language, covering storage, the storage control plane, Inaya Drive, Business Workspace, and Security.
- A complete reference for developers — every public API endpoint, all five published developer packages, and every command across our three command-line tools, documented straight from the real thing, not written from memory.
- Real search, plus a link to the existing AI Docs Assistant for anyone who'd rather just ask a question in plain English.
- Release notes that can never go stale, because they're generated from the same roadmap data the public roadmap page itself uses — one source of truth, not two lists someone has to remember to keep in sync.
- A downloadable technical specification of the API (the industry-standard "OpenAPI" format), so a developer's tools can talk to Inaya's API automatically instead of someone typing it all in by hand.

> **A bonus catch along the way.** Building this also turned up a real, unrelated problem: our live website had actually stopped picking up updates about 20 hours earlier, silently, because of a packaging issue with one internal tool. That's now fixed and confirmed working — every update from this stretch of work, including this documentation site itself, is live on the real site.

> Honestly scoped: this is the first phase of a much larger vision for the documentation site — it doesn't yet have guided tutorials, a full library of solved use cases, or a formal build-checking pipeline. What's live today is real and tested, not a placeholder. Full writeup: docs/official-documentation-platform-report.md.

SECTION 25

Mainframe & Legacy Data Access — Real-Time SQL Virtualization (September 2026)

Inspired by Software AG's CONNX product family: connect an existing relational data source once, publish exactly the tables a company chooses, and query them live with standard SQL — without moving the data or replacing the source system.

- A pluggable connector framework, following the same pattern already proven by Inaya's storage pinning providers — isConfigured/testConnection/discoverMetadata/executeQuery/health/capabilities.
- A real relational reference connector on Node's own node:sqlite: real connection, real metadata discovery, real SQL execution with joins and aggregates.
- A metadata and virtual-schema engine with versioned publishing — a re-import never silently overwrites what was already published.
- A SQL gateway with a real parser, authorization against exactly what has been published, row and timeout limits, and every query audited to the same trail every other Inaya feature uses.
- A real, compiled Java JDBC driver and a real, compiled Windows ODBC driver, so any JDBC- or ODBC-aware tool can connect the same way it would to any other database — registration and connection through the actual Windows ODBC Driver Manager (the path Excel/Power BI use) are now proven, not just the driver in isolation.
- A real RMS/OpenVMS connector over SSH + DCL, proven against a genuine VSI OpenVMS x86-64 instance (7/7 tests against the live system).

What was genuinely not built. Adabas, VSAM and IMS connectors — Adabas has a real, healthy test environment running in Docker, but VSAM/IMS require a genuine z/OS environment, a much larger undertaking. The RMS/OpenVMS connector's fixed-format binary and indexed files aren't supported yet, pending a compiled OpenVMS-side reader. Write-back and federated cross-source joins are deliberately out of scope for this pass. ODBC query execution specifically through the Windows Driver Manager still crashes, isolated to a fault inside Microsoft's own odbc32.dll rather than this driver.

Tested for real: 11 automated tests for the connector/metadata/gateway layer, 7 JDBC integration tests, and 19 ODBC driver tests, all against a live server and real SQLite fixtures. Full writeup: docs/mainframe-legacy-data-access-report.md.

SECTION 26

Sovereign NAS — an On-Premises File Server Managed by Inaya (September 2026)

A control plane and appliance agent that turn a Linux machine on a company's own network into managed, audited file storage: Windows, Mac and Linux computers use it like any office file server (SMB and NFS), while Inaya controls access, snapshots it, watches for ransomware, backs it up, and records every action in the tamper-evident audit trail. It keeps working when the internet is down.

- SMB shares and NFSv4 exports, with recycle bin, quotas, and per-share/per-folder permissions tied to the organization's own membership.
- Storage pools on real RAID1 (or single disks) with Btrfs — a failed disk leaves the pool degraded but writable, and a replacement rebuilds it; a deliberately corrupted block is detected and the read fails rather than returning bad bytes.
- Copy-on-write snapshots, including immutable (WORM) snapshots that root-level deletion cannot defeat, with file or whole-share restore.
- Automatic ransomware detection (mass rewrites, ransom notes, mass deletion, repeated failed logons) that takes an immutable snapshot, alerts, and can apply a time-limited lockdown.
- Backup with file-level dedup, read-back verification, and resumable runs; replication between appliances with manifest verification.

> **The one honest limit.** The supported profile is a Linux VM appliance; it has not been run on physical hardware, so SMART/temperature/UPS data are shown as UNKNOWN rather than invented — the largest remaining gap before selling hardware. Active Directory/LDAP, iSCSI and Kubernetes CSI are not implemented.

> Tested for real: real disk failure injected and recovered, an unclean VM restart with automatic re-attachment, a real Windows SMB client and the Linux kernel NFS client, and a simulated ransomware event with real lockdown and recovery. Full writeup: docs/sovereign-nas-report.md and docs/nas-runbook.md.

SECTION 27

AI Security Workflow — One Checkpoint for Every Text AI Assistant (September 2026)

A single security gateway now sits in front of Inaya's text AI assistants: it detects prompt injection and personal data, applies a per-organization policy, sends risky actions for human approval, and records every decision — replacing six assistants that each used to call the AI model on their own with zero shared safety checks.

- Deterministic detection of five prompt-injection attack families and of personal/sensitive data (email, phone, SSN, Luhn-validated card numbers), with in-place redaction.
- A policy engine with five decisions — allow, warn, redact, block, require human approval — versioned per organization, manager-only to change.
- Coverage of all six text AI routes: the business assistant, wallet assistant, security assistant, Learn assistant, documentation assistant, and OS chat.
- Every decision recorded in the existing tamper-evident audit chain and the Evidence Graph — no second audit system.

> **The one gap, stated plainly.** The voice assistant is not gateway-covered, because speech goes from the browser straight to the speech model, with no server-side text to inspect; it remains authenticated, rate-limited, and re-checks permissions on every tool call. The documentation assistant streams, so only its input is checked, not its output.

> Tested for real: 23 of 23 adversarial tests, plus a real HTTP round-trip against the running app in which a genuine injection attempt on the business assistant was blocked and normal chat was unaffected. Full writeup: AI_SECURITY_IMPLEMENTATION_REPORT.md.

SECTION 28

Native Document & Invoice Automation Engine (September 2026)

One pipeline that turns real Finance, CRM and Procurement records into nine kinds of professional business documents: numbered without gaps, calculated exactly, approved by a second person above a threshold, encrypted and stored, shared through a time-limited or identity-verified link, and independently verifiable by anyone holding the PDF.

- Nine document types — invoice (standard and professional), purchase order, quotation, receipt, customer statement, credit note, debit note, delivery note, business report — sharing one engine, not nine bespoke code paths.
- Exact money handling: minor-unit decimal parsing with the correct exponent for each currency, rounding modes, pro-rata discount allocation.
- A hash-linked evidence chain for every material event, sealed into a portable Document Passport proving Data → Calculation → Template → Document → Approval → Storage → Delivery.
- A safe template language — whitelisted fields, controlled conditions, no code execution, no template injection.

> **Live production defect found and fixed, September 27.** The first real generation on the live site failed: the PDF library's font data was not shipped to the serverless function. Found by this exact live check, diagnosed and fixed the same day, then re-confirmed live in English, Arabic and Urdu, all rendering correctly with proper right-to-left script.

> Tested for real: 59 checks (unit, types, lifecycle, security, end-to-end) against real storage, plus attack scenarios (guessing links, replaying approvals, tampering with stored records, hostile text hidden in an invoice, one organization reaching another's documents). Full writeup: docs/document-invoice-automation-report.md.

SECTION 29

AI Business Operations Manager — Automations (September 2026)

A visual, node-based workflow builder in the Business Workspace: pick a trigger (a schedule, a webhook, an event), add steps that read real Inaya data, an AI Operations Manager that analyses it, rules that decide what happens, and notifications or approval requests — data, then AI, then a decision, then governance, then action, then evidence.

- Triggers: manual, schedule, event, signed webhook, API key, Evidence Graph event, Digital Twin completion, data change.
- Data readers across CRM, overdue invoices, tasks, procurement, inventory, projects, documents, security events, backup status, Trust Health, Business Brief, and — as of this month — Inaya's own support tickets.
- The AI can only call tools it is explicitly given; a tool that changes data can only ever propose, never execute, a change.
- Ready-made templates including Daily Business Health, Invoice Follow-up, Support Escalation, and the new Finance Operations Manager for the AI Bookkeeper.

> **Verified live, not just in tests.** Slack and Gmail delivery were each verified live on 2026-09-26: a real production workflow run posted through a real Slack incoming webhook and sent through the Gmail API, with repeat runs deduplicated so a retry never sends twice.

> Tested for real: unit 16 of 16, security and failure tests 19 of 19, feature tests 12 of 12, the reference-workflow acceptance test 7 of 7, plus a real Gemini structured-output run. A real helpdesk product is not yet verified — only a local stand-in was tested.

SECTION 30

Customer Portal & Customer Service (September 2026)

A complete support desk built on the customer and invoice records Inaya already holds: a customer portal, ticketing with business-hours SLAs, a knowledge base, an AI helper, satisfaction ratings, and an open API — plus Help & Support, so Inaya's own users can raise a ticket with Inaya itself.

- Ticketing with routing, priority policy, business-hours SLAs that correctly pause and resume, saved replies, and internal notes customers never see.
- A responsive customer portal with one-time-link sign-in (and company single sign-on via OpenID Connect with PKCE).
- An AI helper that answers from the company's own knowledge base with citations and hands off to a person when unsure.
- Help & Support: any signed-in Business Workspace member can raise a ticket with Inaya's own support desk, filed from their verified account, with an email copy sent to the support mailbox.

> **The one thing not yet proven with a real provider.** Reply-by-email (a customer's reply becoming a ticket message) is built but depends on the inbound email provider, whose setup is still pending — until then, replying to the support-mailbox copy reaches the customer by ordinary email, but does not post back into the ticket.

> Tested for real: cross-tenant and cross-customer isolation, login-enumeration safety and single-use links, AI outage/retry and prompt-injection handling, CSAT, and API-key scopes. Full writeup: docs/customer-portal-report.md.

SECTION 31

Identity Integration — Entra, Active Directory, SCIM, Rewst and MSP (September 2026)

When a company hires, changes, or dismisses someone in its own directory, Inaya now follows automatically — access is keyed to the directory's own immutable identity, changes are planned/executed/verified, and a leaver's revocation is checked step by step, not assumed.

- Joiner/mover/leaver lifecycle with dry-run, approval, idempotency, and a full evidence trail.
- Standard SCIM 2.0 (Users and Groups) endpoints, plus an optional Microsoft Graph directory pull.
- Six independently verified revocation steps on a leaver — freeze, sessions, credentials, permissions, sharing, break-glass — each with its own PENDING/PARTIAL/COMPLETE/FAILED state and retry.
- Managed-service-provider delegation with two-sided links, four delegated roles, and per-request re-verification.
- A real, standalone on-prem Active Directory sync agent — outbound-only (LDAP to the customer's own domain controller, HTTPS to Inaya, never the reverse), feeding the same signed webhook and engine above.

> **What is proven, and what is not, stated plainly.** Proven against a REAL Microsoft Entra tenant: the Graph directory pull, and a SCIM connection test, joiner and leaver as sent by Entra's own provisioning service — including a leaver with all six revocation steps independently verified. Active Directory is now ALSO proven for real: a genuine Windows Server domain controller was stood up, a real test user created in it, and the new sync agent pulled and pushed her through the unmodified engine end to end. Not yet verified: group-membership push from Entra, Rewst (a paid product, not available to test — Inaya works with any automation platform through its open API), Okta, and HR/PSA/RMM tools.

> Full writeup: docs/identity-integration-report.md, and 14 published documentation pages, each labelled VERIFIED / PARTIAL / UNVERIFIED / UNSUPPORTED / FUTURE.

SECTION 32

AI Bookkeeper (September 2026)

The AI reads bank statements, bills and receipts, works out what each is and which payments and invoices they match, and hands anything doubtful or risky to a person — the AI never posts anything by itself.

- Duplicate-safe bank statement import (CSV/OFX), document ingestion by upload, signed email relay or WhatsApp, with every extracted figure carrying where it came from.
- Payment-to-bill and receipt-to-invoice matching, including part payments, currency differences, one payment covering several invoices, and three-way purchase-order/goods-received/bill matching.
- Configurable confidence policy (default 99%) across four dimensions, with risk always overriding confidence — a large or unusual payment always needs a manager.
- A review queue with ten actions that learns corrections without rewriting history.

> **What is not built, stated plainly.** Inaya has no general ledger and none was invented — "posted" means a recorded payment or a draft expense. Live bank feeds, a live email provider, and a real WhatsApp Business account are not yet verified against real outside accounts; OCR of scanned images depends on the configured AI model and is never auto-processed.

> Tested for real against the live database: flow, unit and security suites, including organization/department isolation, webhook signature and replay protection, and hostile files. Full writeup: docs/ai-bookkeeper-report.md.

SECTION 33

Whole-Codebase Quality & Security Review (September 2026, in progress)

An independent quality review that deliberately attacks the product rather than re-running existing tests: every API route probed for access-control gaps, the real AWS command line and rclone driven against Inaya storage, and the money, bridge and payment flows tested for abuse. Confirmed defects are fixed and locked in with a regression test; the review is ongoing.

- A critical bridge fix: the relayer now signs only messages the source blockchain actually emitted, verified against the real deployed contract — public registration routes could previously have been used to forge a transfer.
- Card payments now settle on-chain exactly once; a retried payment notification previously risked settling twice.
- Storage fixes proven with the real AWS CLI: object names with spaces or accented characters now authenticate, large uploads complete reliably, presigned links work, and the reported file checksum matches the real S3 convention.
- A live production defect (pdfkit's fonts missing from the serverless function, breaking every PDF export) was found by the first live Document Automation test and fixed the same day.

> **Two sweeps now guard the whole product.** One proves no organization can reach another organization's data through any of 453 organization-scoped routes; the other proves no route changes anything for an anonymous visitor, across every remaining route, and that every cron endpoint refuses a wrong secret.

> Still to do: Azure, Google Cloud Storage and Terraform verification with their real tools; mobile, desktop and Inaya Drive on real hardware; concurrency and failure injection; smart-contract static analysis; a complete re-run of the full test suite. Full defect registry: docs/sqa/master-defect-registry.md.

SECTION 34

Managed Database, Document Intelligence & AI/ML Studio (September 2026)

A real, hosted PostgreSQL database provisioned and managed from inside Inaya; a general document-understanding studio that reads any document, not just invoices; and an AI workbench for cataloguing data, checking its quality, registering models, and running one governed piece of code at a time in an isolated, disposable sandbox.

- Databases: provision, start, stop, back up, restore-to-a-point-in-time, and query a real PostgreSQL database through a real external provider (Supabase), all from the Business Workspace — Inaya's own infrastructure cannot run a database engine, so it manages a real one elsewhere rather than pretending to host one itself.
- Document Intelligence: built-in readers for invoices, purchase orders, receipts and contracts, plus a tool to build a custom one for any other document type, each with a version and a status (draft, testing, ready, live) before it is trusted with real work.
- Every extracted value shows its confidence and whether it was actually found in the document's own text — a correction a person makes is added to the record, never silently overwriting what the system originally read.
- AI/ML Studio: a catalogue of an organization's data and models, automatic data-quality checks (missing values, duplicates, out-of-range numbers) against a real data source, and a model registry that fingerprints and version-tracks whatever a data team uploads.
- A single "run this code" button for a data scientist or an AI agent, executed in a brand-new, disposable, internet-off-by-default sandbox that is destroyed the moment it finishes — never a shared, always-on notebook.

> **What this deliberately is not.** Not a persistent notebook (nothing is remembered between two runs) and not a claim that Inaya trains AI models itself — a "model" here is metadata and a file a team already produced elsewhere, tracked and versioned, not trained on Inaya's own infrastructure.

> Tested for real against the live database, a real SQLite-backed data source, and a real, live sandbox created and destroyed on Inaya's own cloud account for each test (26 tests in total, all passing) — including a real bug the tests themselves caught and fixed before anything shipped.

SECTION 35

Post-Quantum Security, a Private Device Cleaner & Private Meetings (October 2026)

A future-proof encryption layer for establishing keys, a local tool that finds junk and duplicate files without ever uploading anything, private peer-to-peer video meetings, and one coherent view over the backup protection Inaya already had — inspired by another privacy-focused product's strongest ideas, built only where Inaya genuinely didn't already have the capability.

- Post-quantum cryptography: a hybrid key-establishment layer combining a classical method with the new NIST-standardized post-quantum method (ML-KEM), so files stay protected even against a future quantum computer — while the actual file encryption itself (AES-256) is unchanged, since that part was never the weak link.
- Inaya Cleaner: scans a computer locally for temporary and duplicate files, never uploads a file inventory anywhere, and moves anything removed to the recycle bin rather than deleting it permanently — reviewed and confirmed by the person before anything happens.
- Inaya Meet: private video meetings where the call itself flows directly between participants' own devices, never through an Inaya server. A host can require approval before someone joins, lock a meeting in progress, and invite guests with no Inaya account at all — including a mode limited to specific invited email addresses, verified with a one-time code.
- Device Protection: backup, sync, local cleanup and device status brought into one place instead of four separate screens, with an at-a-glance badge showing whether each device is ready for the new post-quantum protection.

> **What this deliberately is not.** Not a claim that any part of Inaya is now "quantum-safe" in some generic sense — only the key-establishment step uses the new method, stated exactly that precisely. Not a video-call server that can see meeting content — there's nothing in the middle to see, by design. Not a replacement for the existing backup system — the engineering work here was reconciling and presenting it better, not rebuilding it.

> Verified against a real running server in a real browser — a pass that found and fixed two genuine bugs, including a meeting incorrectly reporting a failure for anyone without a camera, before this was written up. A real two-person video call with real cameras, and a real phone running the mobile app, are the two pieces still waiting on real-world hardware to confirm.

SECTION 36

Secure Collaboration & Enterprise Governance (October 2026)

A Filen/FileCloud-inspired expansion built across 26 workstreams: end-to-end encrypted Secure Chat on the MLS group-messaging standard (RFC 9420) and Secure Notes; Sharing 2.0 with expiring, password-protected and watermarked links plus inbound file requests; enterprise Governance (classification, data-loss prevention, retention, legal hold, a secure document viewer); device trust/block/revoke; and a Sovereign Gateway agent connecting a customer's own on-premises file servers — every feature off by default per organization.

- 114 tracked deliverables across the whole suite, honestly split: 49 finished and independently checked against the real database and, where there's a screen, a real browser; 35 built and tested in code but not yet proven against a real outside system; 27 partly built; 3 not built by deliberate choice (mobile chat and push notifications), not an oversight.
- The Sovereign Gateway, the NTFS/Active-Directory folder-permission bridge, high-availability site replication, and Microsoft 365/Outlook integration are built and tested but not yet proven against a real company file server or a real Microsoft 365 tenant.
- Readiness against the NIST 800-53 control catalogue is self-assessed; no third-party certification is claimed.

> **What this deliberately is not.** The chat protocol's open-source library implementation has not had an independent security audit. This stage stays explicitly in-progress rather than claimed complete — the project's own final verification report says so plainly.

SECTION 37

Sovereign Infrastructure — Bring-Your-Own-Hardware Compute (October 2026)

Inaya could always store and protect files, but had no way to actually run compute for a customer. This closes that gap: a customer connects their own server and it comes online as real, managed infrastructure — virtual machines, block storage, and private networking — on hardware they own, not shared with anyone else, managed from inside the Business Workspace.

- Real QEMU/KVM virtual machines, confirmed by a real guest OS genuinely booting and coming online with its own network address, not just an engine starting and stopping — the first VM in this project to ever do that.
- Real qcow2 block storage with image-backed overlay disks, and a private, isolated network per customer with default-drop firewall rules and NAT.
- A working subset of AWS CLI compatibility (the real, official `aws` command, authenticated with a real Inaya-issued key pair and real SigV4 signing) covering instance, volume and VPC lifecycle — not S3, IAM, or autoscaling.
- A real HAProxy load balancer and real upstream Kubernetes (k3s, not a custom orchestrator), with a working kubeconfig handed to the customer's own `kubectl`.
- Honest TPM and confidential-compute (SEV/TDX) hardware detection — detection only, same precedent as existing GPU detection.

> **What this deliberately is not.** GPU passthrough, full AWS coverage (S3/IAM/autoscaling), HTTP-aware load balancing, and TPM-backed identity or real SEV/TDX execution are explicitly not built — each deferred for a stated, concrete reason (mainly: no dedicated GPU host exists to test real PCI passthrough against), not silently skipped.

> 82+ automated tests pass (30 Rust, 52 Node.js). Real hardware testing caught and fixed 7 genuine bugs a code review alone would have missed, including a pre-existing camelCase/snake_case wire-format mismatch in the node scheduler's capacity comparison and an SSH protocol ordering bug that silently truncated longer guest responses.

SECTION 38

Education OS: Schools and Colleges

A seventh industry mode, for small and mid-sized schools, colleges and academies in Pakistan, India, Bangladesh and the UAE. It runs inside the ERP-style Business Workspace and is built around the three things schools need to keep safe: students' personal data, fee payments, and documents.

- Student and guardian records where phone numbers, home addresses, ID numbers and medical notes are encrypted before they are saved.
- Each staff member sees only what their job needs: the principal and admissions office see the full record, accounts staff see contact details for fee follow-up, and a teacher sees only the students in their own classes, with ID numbers masked.
- Fees in rupees, taka or dirhams: fee invoices for every student in one step, never billed twice for the same month; concessions that need a second person to approve; receipts with running numbers; payments that can be reversed with a reason but never deleted; and a list of who is overdue.
- Admissions, daily attendance, and exams with report cards. Results are locked once the principal publishes them, and any later change is recorded with a reason.
- Parent links: a read-only page for one child's fees, attendance and published results, which expires, can be withdrawn, and never shows ID numbers, medical notes or the address.
- Every opening of a student record, every payment and every mark change goes into the tamper-evident activity record. A student's data can be exported, or erased on request while the fee records are kept.

> **What's live today vs. what's next.** Education OS is built and tested on the web app and the desktop ERP mode. Not built yet: collecting fees online through a payment gateway (fees are recorded, not collected), SMS or WhatsApp fee reminders, timetables, transport and hostel, and mobile screens. We don't claim compliance with India's DPDP Act, the UAE's PDPL, or Pakistani or Bangladeshi data law; the features help a school meet its own obligations.